A GROUPOID WITH FLAT LFP STRUCTURAL MAPS GIVES AN ALGEBRAIC QUOTIENT STACK

THE STACKS PROJECT AUTHORS

Editorial note. Reduction-type proof; both inputs are same-chapter and their roles are displayed. Not a bare pointer: the two conditions are separately addressed.

Allowed standard prerequisites (editorial).

- Bootstrap theorem. theorem-bootstrap (same chapter, named). The criterion to instantiate.
- Flat quotient morphism. lemma-flat-quotient-flat-presentation (same chapter, named). Condition (2) of the criterion.

Source reference. Original lemma 06FH: sources/stacks-criteria/pinned-original-source.tex, source lines 3185–3185. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original theorem 06DC: sources/stacks-criteria/pinned-original-source.tex, source lines 3046–3046. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Theorem 0.1. *Let S be a scheme contained in Sch_{fppf}. Let (U, R, s, t, c) be a groupoid in algebraic spaces over S. Assume s, t are flat and locally of finite presentation. Then the quotient stack $[U/R]$ is an algebraic stack over S.*

Proof. We check the two conditions of Theorem 06DC for the morphism

$$(Sch/U)_{fppf} \longrightarrow [U/R].$$

The first is trivial (as U is an algebraic space). The second is Lemma 06FH. □

Date: stacks-project:criteria.tex:local-snapshot-20261012; GNU FDL 1.2 or later.